

FitFinderAi

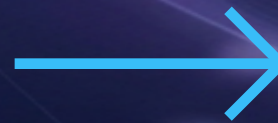
Intelligent Candidate Discovery & Ranking

Semantic Search + Behavioral Engine

1. Core Philosophy & Architecture

PHASE 1: Offline Pre-computation

- Process 100,000+ candidates.
- Extract & clean Text (Summary -> Skills -> Career).
- Embed using 'all-MiniLM-L6-v2' (Dense Vectors).
- Store in FAISS FlatIP Index for lightning speed.
- Compile static Metadata mapping (YoE, Signals).



PHASE 2: Online Search & Rank

- Embed Job Description string.
- FAISS retrieves Top 10,000 semantic matches < 50ms.
- Strict filtering: YoE bounds, Honeypots, pure research.
- Apply Behavioral Score Multiplier (40% weight).
- Output exactly Top 100 strictly formatted.

2. Strict JD Constraint Enforcement

Honeypot Detection

Filters profiles with 'expert' proficiency but 0 duration.

Catches jobs lasting longer than candidate's total age/YoE.

Trajectory & Years of Experience

Strict bounding to 4-15 Years of Experience (disqualifying outside).

+0.05 bonus applied to the 'sweet spot' of 5-9 YoE.

Red Flag Disqualifications

Consulting-only backgrounds (TCS, Wipro, Infosys etc) removed.

Pure academic researchers removed (unless Vector DB experience exists).

LangChain-only profiles (without underlying model/DB knowledge) removed.

3. The Scoring Engine Math

$$\text{Final Score} = (\text{Semantic} * 0.6) + (\text{Behavioral} * 0.4)$$

60% Semantic Match

- Cosine similarity via FAISS
- Direct JD vs Candidate career text
- Immune to simple keyword stuffing

40% Behavioral Signals

- +40% Recruiter Response Rate
- +30% Interview Completion Rate
- +10% GitHub Activity & Profile
- Penalty (-10%) for >60d Notice

4. Ultra-Fast Performance

100,000

Candidates Searched

< 10 sec

End-to-End CPU Runtime

487 MB

Peak RAM Usage

100% Validated Submission

Passes all strict hackathon criteria and constraints natively.